CS471 – Web Technologies (Laboratory)

Lab 1: The Internet Protocols

Student Name: Yasmen Alateeq

Student ID: 432205229

1. Introduction

This lab aims to analyze common internet protocols using Wireshark to understand how data is transmitted across a network. Real network traffic was captured and examined to study the structure and behavior of HTTP, HTTPS, TCP, and UDP protocols.

2. Protocol Analysis

2.1 HTTP (Hypertext Transfer Protocol)

- **Layer:** Application-layer protocol used for transferring web pages.
- **Transport:** Operates over TCP.
- **Port:** Uses default port 80.
- **Security:** Data is transmitted in plain text (unencrypted).
- **Methods:** Common methods include GET and POST.
- **Wireshark Observation:** Easy to analyze as request headers, payload, and response status codes (e.g., 200 OK) are fully visible.

2.2 HTTPS (Hypertext Transfer Protocol Secure)

- **Layer:** Secure version of HTTP.
- **Transport:** Operates over TCP.
- **Port:** Uses default port 443.
- **Security:** Employs TLS/SSL encryption to protect payload and headers.
- **Wireshark Observation:** Raw HTTP content cannot be read. Captured packets appear as encrypted TLS records (e.g., TLSv1.2 / TLSv1.3 Application Data).

2.3 TCP (Transmission Control Protocol)

- **Characteristics:** Connection-oriented and reliable protocol.
- **Handshake:** Establishes connections using a three-way handshake (SYN $\rightarrow$ SYN-ACK $\rightarrow$ ACK).
- **Integrity:** Ensures data integrity and correct ordering using sequence and acknowledgment numbers.
- **Error Handling:** Retransmits lost or corrupted packets.
- **Performance:** Slower than UDP due to management overhead and reliability mechanisms.

- **Applications:** Used by protocols requiring complete reliability, such as HTTP and HTTPS.

2.4 UDP (User Datagram Protocol)

- **Characteristics:** Connectionless and unreliable transport protocol.
- **Handshake:** No handshake sequence or delivery acknowledgments.
- **Performance:** Faster than TCP due to minimal header and processing overhead.
- **Integrity:** Does not guarantee delivery, order, or error recovery.
- **Header Structure:** Minimal 8-byte header consisting of Source Port, Destination Port, Length, and Checksum.
- **Applications:** Commonly used for streaming media, online gaming, and VoIP.

3. Protocol Comparison Tables (Part 4)

Task 1: Protocol Characteristics Comparison

Characteristic	TCP or UDP	Reasons
Reliability and Connection Establishment	**TCP**	TCP is connection-oriented and establishes a formal session using a 3-way handshake (SYN, SYN-ACK, ACK). It guarantees packet delivery via acknowledgments and retransmissions. UDP is connectionless and sends datagrams without confirming receiver readiness or delivery status.
Data Integrity and Ordering	**TCP**	TCP uses sequence numbers to ensure packets are reassembled in the exact order sent and retransmits missing data. UDP delivers datagrams independently without sequence numbers; packets can arrive out of order, duplicated, or dropped without automatic recovery.

Task 2: Use Cases and Performance

Protocol	Use Cases	Performance
TCP	Web browsing (HTTP/HTTPS), file transfers (FTP), email (SMTP/IMAP), and software updates.	**Higher Overhead / Lower Speed:** Requires connection setup, continuous tracking, flow control, and potential retransmission delays.
UDP	Video/audio streaming, VoIP (e.g., Zoom, Discord), real-time online gaming, and DNS queries.	**Lower Overhead / Higher Speed:** Features a lightweight 8-byte header, zero connection latency, and no retransmissions, ideal for real-time applications.

4. Summary of Observations

1. **HTTP Traffic Analysis:** Capturing HTTP traffic on Port 80 revealed clear-text headers, including request methods (GET) and response status codes (200 OK).
2. **TCP Connection Stream:** Following the TCP stream allows viewing the entire bidirectional exchange between client and server.
3. **Encrypted HTTPS Traffic:** Capturing HTTPS traffic showed TLS Application Data records where packet payloads were encrypted and human-unreadable.
4. **Header Comparison:** TCP headers contain complex control information (20+ bytes), while UDP headers remain lightweight (8 bytes), minimizing latency at the cost of delivery guarantees.